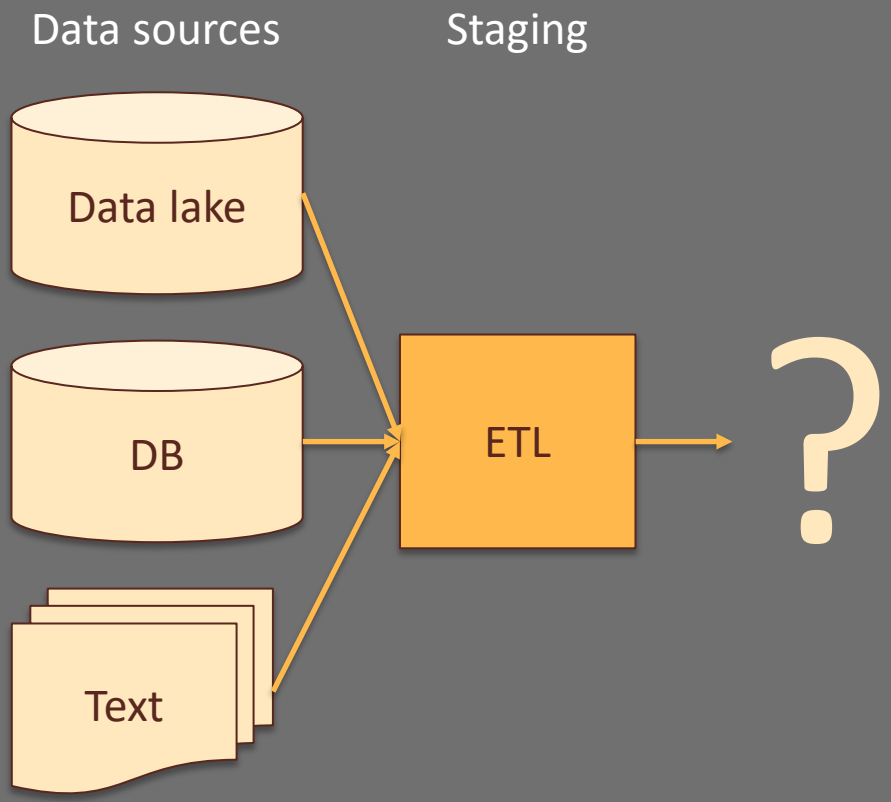
Data Lakes

Main idea

Don’t know everything we want to do with data.

Delay decisions about optimizing storage.

Keep data in original form.


Data sources
Staging
Data lake
DB
Text
ETL
?

Terminology

Structured data
Schema on write

vs

Unstructured data
Schema on read

Metadata

Add metadata (data about data)

- Business
- Operational
- Technical

Demo: Example

https://catalog.data.gov/dataset/201314-integrated-postsecondary-education-data-system

Business metadata

- Description
- Ownership
- Semantic tags
- Policies – e.g., allowed uses, access control
- Limitations – e.g., redactions, anonymization

Operational metadata

- Currency – active, archived, purged
- Lineage – source, history of transformations
- Past usage
- Data quality

Technical metadata

- Size
- Format – CSV, JSON, …
- Schema
- Language
- Version
- Update schedule